Sir Syed University of Engineering & Technology
END SEMESTER EXAMINATIONS FALL 2022

Faculty	Faculty of Computing and Applied Sciences	**Department**	Computer Science and Information Technology	**Batch**	2021F
Semester	3rd	**Degree Program**	BS (Computer Science)	**Course Instructor**	Fariha Anasari/M. Etezaz Ibrahim
Course Code	CS-211	**Course Title**	Discrete Structure	**Credit Hours**	3+0
Date of Examinations	11-02-2023	**Timings**	9:30AM to 12:30PM	**Shift**	Morning

Timings: 3 Hours Max Marks: 50

Instructions:
- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll No.		**Name**		Section	A	**'A' Copy Serial No.**	149262

Q.No.1 **(10 Marks)**

(a) The sum of the series 52+49+46+…= 385. How many terms does the series contain? [5 Marks]
10

(b) A plant grows 1.96 cm in the first week each week it grows by 8% more than it did the week before. By how much does it grow in ten weeks, including the first week? 28.39 [5 Marks]

Q.No.2 **(10 Marks)**

4989600

(a) How many distinct permutation can be made from the letters of the Mathematics? [3 Marks]

(b) How many trees can be planted in a circle if we have 10 trees? 9! [2 Marks]

(c) How many three digits odd number from the digits 0, 1, 3, 4, 5, 7 can be form if each digit can be used only once? $4 \times 4 \times 3$ [5 Marks]
48

Q.No.3 **(10 Marks)**

(a) Calculate gcd (313, 217) by Euclid's algorithm also find whether they are relative prime or not? 1 [5 Marks]

(b) If the product of two integers is $2^5.3^6.5^2.7^{10}$ and their $gcd = 2^3.3^4.5$ what is the lcm? [5 Marks]
$2^2.3^2.5.7^{10}$

Q.No.4 **(10 Marks)**

(a) Consider the Fig 1determine the Pendent Vertices, Pendent Edges, Odd and Even Vertices.

[5 Marks]

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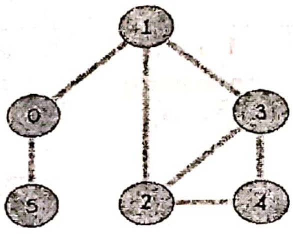

Fig. 1

PV = 0
Pendent edge 5
odd = 1, 2, 3
even = 4, 5, 0

(b) Define Euler path, Euler circuit and Euler graph with any suitable small graph. [5 Marks]

Q.No.5	(10 Marks)

(a) Find preorder, inorder and post order in tree traversal by the given Fig 2 [5 Marks]

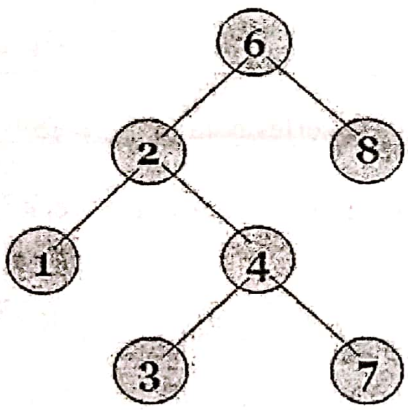

Fig 2

In = 6, 2, 1, 4, 3, 7, 8
Pre = 1, 2, 3, 4, 7, 6, 8
Post = 1, 3, 7, 4, 2, 8, 6

(b) Let a connected simple planner graph has 30 vertices, each of degree 4. How many regions does a representative of this planner graph spilt the plane? [5 Marks]

$\gamma = 32$ $e = 60$